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RADIATION DETECTOR STRUCTURES WITH SIDEWALL COATING AND METHODS OF FABRICATION THEREOF

Abstract

A detector structure includes a carrier board, at least one application specific integrated circuit (ASIC) located over the carrier board, the at least one ASIC including signal processing channel circuitry, at least one radiation sensor located over a front side of the at least one ASIC, and a protective coating located over sidewalls of the at least one radiation sensor and over at least a portion of the carrier board and the at least one ASIC. In various embodiments, the detector structure including the at least one radiation sensor is assembled prior to depositing a protective coating layer over portions of the detector structure including the sidewalls of the at least one radiation sensor.

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Background/Summary

RELATED APPLICATION [0001] This application claims the benefit of priority to U.S. Provisional Application No. 63/563,076, filed on Mar. 8, 2024, which is incorporated by reference herein in its entirety.

FIELD

[0002] The present disclosure relates generally to radiation detectors, and more specifically to radiation detector structures having a sidewall coating and methods of forming a sidewall coating over a radiation detector structure.

BACKGROUND

[0003] Room temperature pixelated radiation detectors made of semiconductors, such as cadmium zinc telluride ($\text{Cd}_{1-x}\text{Zn}_x\text{Te}$ where $0 < x < 1$, or “CZT”), are gaining popularity for use in medical and non-medical imaging. These applications use the high energy resolution and sensitivity of the radiation detectors.

SUMMARY

[0004] According to an aspect of the present disclosure, a detector structure includes a carrier board, at least one application specific integrated circuit (ASIC) located over the carrier board, the at least one ASIC including signal processing channel circuitry, at least one radiation sensor located over a front side of the at least one ASIC, the at least one radiation sensor having a front side and a back side, wherein the back side of the at least one radiation sensor faces the front side of the at least one ASIC; and a protective coating located over sidewalls and at least a portion of the back side of the at least one radiation sensor and over at least a portion of the at least one ASIC and over at least a portion of the carrier board.

[0005] According to another aspect of the present disclosure, a method of fabricating a detector structure includes assembling a detector structure by mounting at least one ASIC over a front side of a carrier substrate and at least one radiation sensor over a front side of the at least one ASIC, and forming a protective coating over at least a portion of the assembled detector structure including over sidewalls of the at least one radiation sensor.

[0006] Further embodiments include X-ray imaging systems including a radiation source configured to emit an X-ray beam, and a plurality of above-described detector structures that form a continuous detector surface and that are configured to receive the X-ray beam from the radiation source through an intervening space configured to contain an object therein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. 1A is a functional block diagram of an X-ray imaging system in accordance with various embodiments of the present disclosure.

[0008] FIG. 1B is a schematic illustration of an application specific integrated circuit (ASIC) configured to count X-ray photons detected in each pixel detector within a set of energy bins according to various embodiments of the present disclosure.

[0009] FIG. 2A is a rear perspective view of a detector array for a computed tomography (CT) X-ray imaging system according to various embodiments of the present disclosure.

[0010] FIG. 2B is a perspective view of a CT X-ray imaging system illustrating the orientation of the detector array with respect to an X-ray source and a patient being imaged according to various embodiments of the present disclosure.

[0011] FIG. 3A is a vertical cross-sectional view of a radiation detector unit according to one embodiment of the present disclosure.

[0012] FIG. 3B is a side view of an alternative configuration of a radiation detector unit according to various embodiments of the present disclosure.

[0013] FIG. 4 is a perspective view of a detector module including a plurality of radiation detector units mounted to a frame bar according to various embodiments of the present disclosure.

[0014] FIG. 5 is a top view of a detector array including a plurality of detector modules according to various embodiments of the present disclosure.

[0015] FIG. 6A is a side cross-section view of a radiation detector unit including a carrier board, an ASIC mounted over the front side of the carrier board, and a plurality of radiation sensors mounted over the front side of the ASIC according to various embodiments of the present disclosure.

[0016] FIG. 6B is a side cross-section view of the radiation detector unit with a mask formed over select portions of the radiation detector unit according to various embodiments of the present disclosure.

[0017] FIG. 6C is a side cross-section view of the radiation detector unit with a protective coating deposited over exposed surfaces of a radiation detector unit according to various embodiments of the present disclosure.

[0018] FIG. 6D is a side cross-section view of the radiation detector unit with a protective coating following the removal of the mask according to various embodiments of the present disclosure.

[0019] FIG. 6E is a side cross-section view of a radiation sensor according to an alternative embodiment.

[0020] FIG. 7 is a side cross-section view of the radiation detector unit with a protective coating according to another embodiment of the present disclosure.

[0021] FIG. 8 is a flow diagram illustrating a method of fabricating a detector structure according to various embodiments of the present disclosure.

DETAILED DESCRIPTION

[0022] Embodiments of the present disclosure provide detector arrays for ionizing radiation, the various aspects of which are described herein with reference to the drawings.

[0023] The various embodiments will be described in detail with reference to the accompanying drawings. Wherever possible, the same reference numbers will be used throughout the drawings to refer to the same or like parts. References made to particular examples and implementations are for illustrative purposes, and are not intended to limit the scope of the invention or the claims. Any reference to claim elements in the singular, for example, using the articles “a,” “an,” or “the” is not to be construed as limiting the element to the singular. The terms “example,” “exemplary,” or any term of the like are used herein to mean serving as an example, instance, or illustration. Any implementation described herein as an “example” is not necessarily to be construed as preferred or advantageous over another implementation. The drawings are not drawn to scale. Multiple instances of an element may be duplicated where a single instance of the element is illustrated, unless absence of duplication of elements is expressly described or clearly indicated otherwise.

[0024] FIG. 1A is a functional block diagram of an X-ray imaging system **100** in accordance with various embodiments. The X-ray imaging system **100** may include an X-ray source **110** (i.e., a source of ionizing radiation), and an energy discriminating photon counting radiation detector array **300**. The X-ray imaging system **100** may additionally include a patient support structure **105**, such as a table or frame, which may rest on the floor and may support an object **10** to be scanned. In some embodiments, the object **10** may be a biologic subject (i.e., a human or animal patient). The support structure **105** may be stationary (i.e., non-moving) or may be configured to move relative to other elements of the X-ray imaging system **100**, such as the X-ray source. For example, in a non-destructive testing imaging system, the support structure **105** may comprise a moving belt or web which supports the object **10**. The object **10** may laterally translated by the moving belt or web past a stationary X-ray source **110**, while the stationary detector array **300** is located under the belt or web. The object **10** may comprise food, baggage, manufactured products, or any other object that is subject to non-destructive testing to determine the object's composition (e.g., whether the

food contains impurities), contents (e.g., contents of baggage), and/or defects (e.g., defects in manufactured products).

[0025] The X-ray source **110** may be mounted to a gantry (e.g., for CT imaging) or another support, and may move or remain stationary (e.g., in non-destructive testing) relative to the object **10**. The X-ray source **110** is configured to deliver ionizing radiation to the radiation detector array **300** by emitting an X-ray beam **107** toward the object **10** and the radiation detector array **300**. After the X-ray beam **107** is attenuated by the object **10**, the beam of radiation **107** is received by the radiation detector array **300**. In an alternative mode, the X-ray beam is diffracted by the object **10**, the beam of radiation **107** is received by the radiation detector array **300**.

[0026] The radiation detector array **300** may include one or more radiation sensors **80** coupled to detector read-out circuitry **130**. Each radiation sensor **80** may be controlled by a high voltage bias power supply **124** that selectively creates an electric field between an anode **128** and cathode **122** pair coupled thereto. In one embodiment, each radiation sensor **80** includes a plurality of anodes **128** (e.g., one anode per pixel) and one common cathode **122** electrically connected to the power supply **124** and facing the X-ray source **110**. Each radiation sensor **80** may include a detector material **125**, such as a semiconductor material disposed between the anode **128** and cathode **122** and thus configured to be exposed to the electrical field therebetween. The semiconductor material may comprise any suitable semiconductor material for detecting X-ray radiation disposed between the anode **128** and cathode **122** and thus configured to be exposed to the electrical field therebetween. In various embodiments, the semiconductor material of the radiation sensor(s) **80** may comprise a II-VI semiconductor material, such as cadmium telluride, cadmium zinc telluride (i.e., CdZnTe or “CZT”), cadmium selenide telluride, and cadmium zinc selenide telluride. Other suitable semiconductor materials are within the contemplated scope of disclosure. These materials may include a “High Z” element.

[0027] The detector read-out circuitry may include one or more application specific integrated circuits (ASICs) **130**. Each ASIC **130** may be coupled to one or more radiation sensors **80** and may receive signals (e.g., charge or current) from the anodes **128** of the radiation sensor(s) **80**.

Alternatively, multiple ASICs **130** may be coupled to one radiation sensor **80**. Each ASIC **130** may be configured to provide data to and by controlled by a control unit **170**. Each of the radiation sensors **80** may be segmented or configured into a large number of small “pixel” detectors **126**. In various embodiments, the pixel detectors **126** of the radiation sensors **80** and the ASIC(s) **130** are configured to output data that includes counts of photons detected in each pixel detector **126** in each of a number of energy bins. Thus, radiation detector arrays **300** of various embodiments may provide both two-dimensional detection information regarding where photons were detected, thereby providing image information, and measurements of the energy of the detected X-ray photons. A radiation detector array **300** that is capable of measuring the energy of the X-ray photons impinging on the array **300** may be referred to as an energy-discriminating radiation detector array **300**.

[0028] The control unit **170** may be configured to synchronize the X-ray source **110**, the read-out ASIC(s) **130**, and the high voltage bias power supply **124**. The control unit **170** may be coupled to and operated from a computing device **160**. Alternatively, the computing device **160** and the control unit **170** may be integrated together as one device.

[0029] In some embodiments, the X-ray imaging system **100** may be a computed tomography (CT) imaging system. The CT imaging system **100** may include a gantry (not shown in FIG. 1A), which may include a moving part, such as a circular, rotating frame with the X-ray source **110** mounted on one side and the radiation detector array **300** mounted on the other side. The radiation detector array **300** may have a curved shape along its long axis (i.e., the x-axis direction in FIG. 1A) such that each of the pixel detectors **126** along the length of the radiation detector may face towards the focal spot of the X-ray source **110**. The gantry may also include a stationary (i.e., non-moving) part, such as a support, legs, mounting frame, etc., which rests on the floor and supports the moving

part. The X-ray source **110** may emit a fan-shaped or cone-shaped X-ray beam **107** as the X-ray source **110** and the radiation detector array **300** rotate on the moving part of the gantry around the object **10** to be scanned. After the X-ray beam **107** is attenuated by the object **10**, the X-ray beam **107** is received by the radiation detector array **300**. The curved shape of the radiation detector array **300** may allow the CT imaging system **100** to most effectively reject radiation scattered by the object **10**.

[0030] For each complete rotation of the X-ray source **110** and the radiation detector array **300** around the object **10**, one cross-sectional slice of the object **10** may be acquired. As the X-ray source **110** and the radiation detector array **300** continue to rotate, the radiation detector array **300** may take numerous snapshots called “views”. The X-ray source **110** and the detector array **300** may slowly move relative to the object (e.g., patient) **10** along a horizontal direction (i.e., into and out of the page in FIG. 1A) so that the detector array **300** may capture incremental cross-sectional profiles over a region of interest (ROI) of the object **10**, which may include the entire object **10**. The data acquired by the radiation sensor(s) **80** and output by the read-out ASIC(s) **130** may be passed along to the computing device **160** that may be located remotely from the radiation detector array **300** via a connection **165**. The connection **165** may be any type of wired or wireless connection. If the connection **165** is a wired connection, the connection **165** may include a slip ring electrical connection between any structure (e.g., gantry) supporting the radiation detector array **300** and a stationary support part of the support structure, which supports any part (e.g., a rotating ring). If the connection **165** is a wireless connection, the radiation detector array **300** may contain any suitable wireless transceiver to communicate data with another wireless transceiver that is in communication with the computing device **160**. The computing device **160** may include processing and imaging applications that analyze each profile obtained by the radiation detector array **300**, and a full set of profiles may be compiled to form a three-dimensional computed tomographic (CT) reconstruction of the object **10** and/or two-dimensional images of cross-sectional slices of the object **10**.

[0031] Various alternatives to the design of the X-ray imaging system **100** of FIG. 1A may be employed to practice embodiments of the present disclosure. X-ray imaging systems may be designed in various architectures and configurations. For example, an X-ray imaging system may have a helical architecture. In a helical X-ray imaging scanner, the X-ray source **110** and radiation detector array **300** are attached to a freely rotating gantry. During a scan, the support structure (e.g., table) **105** moves the object **10** smoothly through the scanner, or alternatively, the X-ray source **110** and detector array **300** may move along the length of the object **10**, creating helical path traced out by the X-ray beam. Alternatively, in an orbital X-ray imaging scanner, the object **10** may remain stationary while the gantry rotates completely or partially around the object **10** (e.g., to image a heart or a brain in one revolution). Slip rings may be used to transfer power and/or data on and off the rotating gantry. In other embodiments, the X-ray imaging system may be a tomosynthesis X-ray imaging system. In a tomosynthesis X-ray scanner, the gantry may move in a limited rotation angle (e.g., between 15 degrees and 60 degrees) in order to detect a cross-sectional slice of the object **10**. The tomosynthesis X-ray scanner may be able to acquire slices at different depths and with different thicknesses that may be reconstructed via image processing. Alternatively, the X-Ray imaging system **100** may be a non-destructive testing system.

[0032] FIG. 1B illustrates components of an X-ray imaging system, including components within the ASIC **130** configured to count X-ray photons detected in each pixel detector **126** within a set of energy bins. As used herein, the terms “energy bin” and “bin” refer to a particular range of measured photon energies between a minimum energy threshold and a maximum energy threshold. For example, a first bin may refer to counts of photons determined to have an energy greater than a threshold energy (referred to as a trigger threshold, e.g., 20 keV) and less than 40 keV, while a second bin may refer to counts of photons determined to have an energy greater than 40 keV and less than 60 keV, and so forth.

[0033] X-rays **107** from an X-ray source (e.g., X-ray tube) **110** may be attenuated by a target (e.g., an object **10**, such as a human or animal patient) before interacting with the radiation detector material within the pixelated detector array **300**. An X-ray photon interacting (e.g., via the photoelectric effect) with a pixelated radiation detector material generates an electron cloud within the material that is swept by an electric field to the anode electrode **128**. The charge gathered on the anode creates a signal that is integrated by a charge sensitive amplifier (CSA) **131**. There may be a CSA **131** for each pixel detector **126** (e.g., for each anode **128**) within the pixelated X-ray detector array **300**. The voltage of the CSA output signal may be proportional to the energy of the X-ray photon. The output signal of the CSA may be processed by an analog filter or shaper **132**.

[0034] The filtered output may be connected to the inputs of a number of analog comparators **134**, with each comparator connected to a digital-to-analog converter (DAC) **133** that inputs to the comparator a DAC output voltage that corresponds to the threshold level defining the limits of an energy bin. The detector circuitry **130** may be configured so that after the CSA voltage has stabilized (after the dead time), that voltage may be between two voltage thresholds set by two DACs **133**, which determines the output of the comparators **134**. Outputs from the comparators **134** may be processed through decision gates **137**, with a positive output from a comparator **134** corresponding to a particular energy bin (defined by the DAC output voltages) resulting in a count added to an associated counter **135** for the particular energy bin. Periodically, the counts in each energy bin counter **135** are output as signals **138** to the control unit **170**.

[0035] Other suitable configurations for the read-out electronics of the ASIC **130** are within the contemplated scope of disclosure. For example, in some configurations, the analog voltage signals from the CSA may be converted to digital signals using an analog-to-digital converter (ADC) prior to being sorted into the respective energy bins.

[0036] The detector array **300** of an X-ray imaging system may include an array of radiation detector elements, referred to herein as pixel detectors. The signals from the pixel detectors may be processed by a pixel detector circuit (e.g., an above-described ASIC **130**), which may sort detected photons into energy bins based on the energy of each photon or the voltage generated by the received photon. When an X-ray photon is detected, its energy is determined and the X-ray photon count for its associated energy bin is incremented. For example, if the detected energy of an X-ray photon is 24 kilo-electron-volts (keV), the X-ray photon count for the energy bin of 20-40 keV may be incremented. The number of energy bins may be three or more, such as four to twelve. In an illustrative example, an X-ray photon counting detector may have four energy bins: a first bin for detecting photons having an energy between 20 keV and 40 keV, a second bin for detecting photons having an energy between 40 keV and 60 keV, a third bin for detecting photons having an energy between 60 keV and 90 keV, and a fourth bin for detecting photons having an energy above 90 keV (e.g., between 90 keV and 120 keV). The greater the total number of energy bins, the better the material discrimination. The total number of energy bins and the energy range of each bin may be selectable by a user, such as by adjusting the threshold levels defining the limits of the respective energy bins in the read-out ASIC **130** as shown in FIG. 1B.

[0037] In various embodiments, a detector array **300** for an X-ray imaging system **100** as described above may include a plurality of pixel detectors **126** extending over a two-dimensional (2D) detector array surface. A typical radiation detector array **300** may include an array of individual radiation sensors **80** arranged side-by-side to provide the 2D detector array surface. The radiation sensors **80** may be located sufficiently close to each other to treat the 2D detector surface as essentially continuous, even though there are small gaps present between adjacent radiation sensors **80**. Each radiation sensor **80** may comprise a semiconductor detector material plate **125**, a continuous cathode electrode **122** located on a first side of the semiconductor detector material plate **125**, and a plurality of anode electrodes **128** located on a second side of the semiconductor detector material plate **125**. Each of the pixel detectors **126** comprises one the plurality of anode electrodes **128** and portions of the continuous cathode electrode **122** and the semiconductor

detector material plate **125** overlying the one of the plurality of anode electrodes **128**.

[0038] The detector array **300** (which is also referred herein as a detector module system (DMS)) may further include a modular configuration including a plurality of detector modules, where each detector module may include at least one above-described radiation sensor **80**, at least one ASIC **130** (also known as a read-out integrated circuit (ROIC)) electrically coupled to the at least one radiation sensor **80**, and a module circuit board. The module circuit board may support transmission of electrical power, control signals, and data signals between the module circuit board and the at least one ASIC **130** and the at least one radiation sensor **80** of the detector module, and may further support transmission of electrical power, control signals, and data signals between the module circuit board and the control unit **170** of the X-ray imaging system **100**, other module circuit boards of the detector array, and/or a power supply for the detector array. A plurality of detector modules may be assembled on a common support structure, such as a detector array frame, to form a detector array **300**.

[0039] FIG. 2A is a rear perspective view of a detector array **300** for a computed tomography (CT) X-ray imaging system according to various embodiment of the present disclosure. The detector array **300** in this embodiment includes multiple detector modules **200** mounted on a detector array frame **310**. The detector array frame **310** may be configured to provide attachment of a row of detector modules **200** such that physically exposed surfaces of the radiation sensors **80** of the detector modules **200** collectively form a curved detection surface located within a cylindrical surface. The multiple detector modules **200** may be assembled such that radiation sensors attached to neighboring detector modules **200** abut each other, i.e., make direct surface contact with each other and/or include a gap between adjacent radiation sensors that is less than 3 mm, and/or less than 2 mm, and/or less than 1 mm in the x-direction. In some embodiments, the detector modules **200** may be mounted to the detector array frame **310** by attaching frame bars **140** of the detector modules **200** to the detector array frame **310** using suitable mechanical fasteners. The radiation sensors and ASICs **130** of each module **200** may be mounted over a first (i.e., front) surface of the frame bar **140**. Each module **200** may also include a module circuit board **220** extending away from a rear surface of the frame bar **140**. Major surfaces of the module circuit boards **220** of the detector modules **200** may face each other in the detector array **300**.

[0040] FIG. 2B is a perspective view of a CT X-ray imaging system **100** illustrating the orientation of the detector array **300** with respect to an X-ray source **110** and an object (e.g., patient) **10** being imaged according to various embodiments of the present disclosure. Referring to FIG. 2B, the X-ray source **110** and the detector array **300** (e.g., DMS) may rotate around the object **10** and the support structure (e.g., motorized table) **105** along the direction of arrow **306** to obtain cross-sectional image profiles (or “slices”) of the object **10**. The X-ray source **110** and the detector array **300** may also be translated relative to the object **10** (e.g., by moving the support structure **105** and the object **10** with respect to the X-ray source **110** and the detector array **300** and/or by moving the X-ray source **110** and the detector array **300** along the length of the object **10**) along a horizontal direction to obtain cross-sectional image “slices” of different portions of the object **10**. The direction of the horizontal movement of the X-ray source **110** and the detector array **300** relative to the object **10** may be referred to as the “Z-axis” direction, which may be parallel to the axis of rotation of the X-ray source **110** and the detector array **300** around the object **10**. As discussed above, the detector array **300** may also have a curved shape along the direction in which the X-ray source **110** and the detector array **300** rotate around the object **10**. The pixel detectors **126** of the detector array **300** may be arranged in multiple columns and rows of pixel detectors, where each column may extend along the Z-axis direction, and each row may extend along the direction of rotation **306** (e.g., angular Φ direction) of the detector array **300** around the object **10**. Accordingly, the location of each pixel detector **126** within the detector array **300** may be defined by a unique row and column pair, where the location of the pixel detector **126** within a given column may be defined by its location along the Z-axis direction, and the location of the pixel detector within a

given row may be defined by the azimuth angle Φ of a line segment extending between the pixel detector **126** and the focal spot of the X-ray source **110**, where all pixel detectors **126** within the same column may have the same azimuth angle Φ . The detector array **300** shown in FIG. 2B may be similar to the detector array **300** described above with reference to FIG. 2A. The detector array **300** may further include a suitable housing or enclosure **305** that encloses and protects the module circuit boards **220**.

[0041] In some embodiments, each of the detector modules **200** of the detector array **300**, such as the detector array **300** shown in FIGS. 2A and 2B may be constructed from a set of radiation detector units, which may also be referred to as “mini-modules” or “submodules.” In some embodiments, each of the radiation detector units may include one or more radiation sensors **80** coupled to a single ASIC **130**. FIG. 3A is a vertical cross-sectional view of a radiation detector unit **210** according to one embodiment of the present disclosure. Referring to FIG. 3A, the radiation detector unit **210** includes a pair of above-described radiation sensors **80**, an ASIC **130**, and an interposer **40** disposed between the radiation sensors **80** and the ASIC **130**. The interposer **40** includes an insulating interposer matrix **44**, which may include semiconductor, glass, polymer (e.g., printed circuit board insulating laminate) or ceramic material, and a plurality of metal interconnect structures **42** embedded within the insulating interposer matrix **44**. Bonding pads (not expressly shown) may be located on the front side and the backside of the interposer **40** and may be electrically coupled to the metal interconnect structures **42**. As used herein, the “front side” of elements refers to the side that faces the incoming radiation, and the “backside” of elements refers to the side that is the opposite side of the front side.

[0042] The ASIC **130** may interface with external components through bonding pads that are located on the front side of the ASIC **130**. The bonding pads may include input pads, output pads, and power pad. The bonding pad(s) of the ASIC **130** can be arranged as an array, such as a rectangular array. At least a portion of the bonding pads on the backside of the interposer **40** may have the same periodicity as the bonding pads on the front side of the ASIC **130**.

[0043] Each of the radiation sensors **80** may have bonding pads located on the backside of the radiation sensor **80**. The bonding pads on the backside of the radiation sensors **80** may be arranged as an array, such as a rectangular array. The front side bonding pads of the interposer **40** may have the same periodicity as the periodicity of bonding pads on the backside of the radiation sensors **80**.

[0044] Referring again to FIG. 3A, the ASIC **130** may be mounted to the backside of the interposer **40** through an array of first bonding structures **32**, such as solder balls or copper pillars.

Specifically, the array of first bonding structures **32**, may be bonded to a respective array of a bonding pads (not shown for clarity) on the front side of the ASIC **130** and an array of backside bonding pads of the interposer **40** employing a flip-chip bonding process (e.g., a C4 bonding process and/or a thermo-compression process in embodiments using copper pillar bonding structures). An insulating matrix **34** may be formed around the array of first bonding structures **32** to structurally reinforce the array of first bonding structures **32**. While a configuration in which one ASIC **130** is bonded to the backside of the interposer **40** is illustrated herein, two or more ASICs **130** may be bonded to the backside of the interposer **40** in some embodiments. At least one radiation sensor **80** may be bonded to the front side of the interposer **40** via bonding material portions **82**. In some embodiments, the bonding material portions **82** may include a low temperature solder material or conductive epoxy. In one embodiment, the at least one radiation sensor **80** includes a pair of radiation sensors **80** having a respective rectangular shape and adjoined to each other with no gap or with a gap less than 3 mm, and/or less than 2 mm, and/or less than 1 mm. X-ray photon detection signals from the radiation sensors **80** may be transmitted to the ASIC **130** via the interposer **40**. The ASIC **130** may be configured to convert event detection signals from the at least one radiation sensor **80** to digital detection signals, which can include the pixel location and the energy range of the detected radiation.

[0045] The radiation detector unit **210** may further include a carrier board **60** and at least one flex

cable assembly **62**, which are configured to route power supply to the ASIC **130** and to the at least one radiation sensor **80**, control signals to the ASIC **130**, and data signals (e.g., digital detection signals) generated by the ASIC **130** and transmitted through the interposer **40**. One end of each flex cable assembly **62** may be attached to a respective side of the carrier board **60**, and another end of each flex cable assembly may be connected to a module circuit board **220** as shown in FIG. 2. The carrier board **60** may be a printed circuit board including an insulating substrate and printed interconnection circuits. A thermally conductive paste **36**, such as a silver paste layer, may be provided between the backside of the ASIC **130** and the front side of the carrier board **60**. An array of second solder balls **52** may each be bonded to a bonding pad **46** on the backside of the interposer **40** and a corresponding bonding pad (not expressly shown in FIG. 3A) on the front side of the carrier board **60**. The second solder balls **52** may be located around and laterally spaced away from peripheral side surfaces of the ASIC **130**. Each flex cable assembly **62** may include signal wires and power wires. The signal wires are configured to transmit the electronic detection signals and control signals, and the power wires are configured to provide electrical power to the interposer **40**, which distributes the electrical power to the ASIC **130** and the radiation sensor(s) **80**. In some embodiments, the flex cable assembly **62** can be more flexible (i.e., can bend with a lower application of force) than the carrier board **60**.

[0046] The radiation sensors **80**, the interposer **40**, the ASIC **130** and the carrier board **60** may be mounted to a supporting substrate (e.g., a block) **90** as shown in FIG. 3A. The supporting substrate may include a high thermal conductivity material such as a metal (e.g., aluminum, copper, etc.). The supporting substrate **90** may function as a heat sink for the radiation detector unit **210**. The supporting substrate **90** may be attached to the backside of the carrier board **60** using a thermally conductive adhesive such as a thermally conductive paste, and/or by mechanical connection structures (such as snap-in connectors, screws, and/or bolts and nuts).

[0047] FIG. 3B is a vertical cross-section view of an alternative configuration of a radiation detector unit **210** according to an alternative embodiment of the present disclosure. In this embodiment, a pair of radiation sensors **80** is directly mounted to an ASIC **130** via a plurality of bonding material portions **82**. Referring to FIG. 3B, each radiation sensor **80** may include an above-described detector material **125** having at least one cathode electrode **122** on a front side of the radiation sensor **80** and a plurality of anode electrodes **128** on a back side of the radiation sensor **80** defining an array of pixel detectors **126** as described above. Each radiation sensor **80** may be directly mounted to the front side of the ASIC **130** via a plurality of bonding material portions **82**. In other words, each radiation sensor **80** may be mechanically and electrically coupled to the ASIC **130** via the plurality of bonding material portions **82**, and no interposer **40** or similar intervening structural component for routing of electrical signals between the radiation sensors **80** and the ASIC **130** is located between the back side of each of the radiation sensors **80** and the front side of the ASIC **130**. Such a configuration may be referred to as a “direct attach” radiation detector unit **210**. Exemplary embodiments of “direct attach” radiation detector units **210** and detector modules **200** are described, for example, in U.S. Provisional Patent Application No. 63/380,769, filed on Oct. 25, 2022, and U.S. patent application Ser. No. 18/158,695, filed on Jan. 24, 2023, the entire teachings of both of which are incorporated by reference herein for all purposes.

[0048] The plurality of bonding material portions **82** may be arranged in an array, such as a rectangular array, having the same periodicity as the periodicity of the anode electrodes **128** on the back sides of the radiation sensors **80**. Thus, each bonding material portion **82** may electrically couple a respective anode electrode **128** of a radiation sensor **80** to the front side of the ASIC **130**. In one non-limiting embodiment, the bonding material portions **82** may be composed of a conductive epoxy. Other suitable bonding materials, such as a low temperature solder material coupled with under bump metallization on the anode electrodes **128**, may be utilized to mount the radiation sensor **80** to the front side of the ASIC **130**. An optional underfill material (not shown in FIG. 3B) may be provided in the space between the back side surface of the radiation sensors **80**

and the front side surface of the ASIC **130** and laterally surrounding the bonding material portions **82**. The underfill material may include a suitable insulating material, such as an insulating epoxy material or an inorganic dielectric material, such as TiO₂ or mixtures thereof

[0049] In the embodiment illustrated in FIG. **3B**, a pair of radiation sensors **80** are mounted to the front side of the ASIC **130**, although it will be understood that in other embodiments, a single radiation sensor **80** or more than two radiation sensors **80** may be mounted to the front side of the ASIC **130**. In some embodiments, the horizontal dimensions of the ASIC **130** may be substantially equal to or slightly smaller than the combined dimensions of the one or more radiation sensors **80** mounted thereto along the corresponding horizontal directions. Thus, each pixel detector **126** of the radiation sensor(s) **80** may overlie a corresponding pixel region **180** of the ASIC **130**. In some embodiments, the ASIC **130** and each of the radiation sensors **80** mounted thereto may have a rectangular periphery. This may enable any of the four peripheral sides of the radiation detector unit **210** to be abutted against a peripheral side of an adjacent radiation detector unit **210** upon assembly of multiple radiation detector units **210** in a detector array **300**.

[0050] The radiation detector unit **210** of FIG. **3B** may further include a carrier board **60** that is configured to route power supply to the ASIC **130** and to the at least one radiation sensor **80**, control signals to the ASIC **130**, and data signals (e.g., digital detection signals) generated by the ASIC **130**. One or more cables **62**, such as a flex cable assembly, may be attached to a respective side of the carrier board **60**, and another end of each cable may be connected to the above-described module circuit board **220**. The carrier board **60** may be a printed circuit board including an insulating substrate and printed interconnection circuits. In various embodiments, the ASIC **130** may be disposed over the carrier board **60** such that the back side of the ASIC **130** may contact the front side of the carrier board **60**.

[0051] Optionally, a plurality of through-substrate vias (TSVs) **190** may be provided in the ASIC **130**. The TSVs **190** may include an electrically conductive material (e.g., a metal material, such as copper) that extends between the front side and the back side of the ASIC **130**. In embodiments in which the ASIC **130** may be formed on and/or in a silicon substrate, the TSVs **190** may also be referred to as “through-silicon vias.”

[0052] Accordingly, electrical connections between the carrier board **60** and the ASIC **130** may be made through the back side of the ASIC **130** via the plurality of TSVs **190**. In particular, each of the TSVs **190** may electrically contact a conductive trace **191** located on the front side of the carrier board **60**, as schematically illustrated in FIG. **3B**. This may obviate the need for wire bond and/or interposer connections between the front side of the carrier board **60** and the front side of the ASIC **130**, which may help to minimize the footprint of the radiation detector unit **210**. In some embodiments, the TSVs **190** may be connected to the metal interconnect features **191** on the carrier board **60** via a plurality of bonding material portions **192**, which may include, for example, solder balls. In various embodiments, outer periphery of the carrier board **60** may not extend beyond the outer periphery of the ASIC(s) **130** and radiation sensor(s) **80** located over the carrier board **60** so as to provide a radiation detector unit **210** that is buttable on all four sides.

[0053] In other embodiments, a portion of the ASIC **130** may extend beyond the outer periphery of the radiation sensor(s) **80** and a portion of the carrier board **60** may extend beyond the outer periphery of the ASIC **130**. A plurality of wire bond connections may extend between the front side of the carrier board **60** and the front side of the ASIC **130**, as is described and illustrated in the above-referenced U.S. patent application Ser. No. 18/158,695. The wire bond connections between the carrier board **60** and the ASIC **130** may be in addition to, or may be in lieu of, electrical connections between the front side of the carrier board **60** and the back side of the ASIC **130** via TSVs **190** as described above.

[0054] FIG. **4** is a perspective view of a detector module **200** including a plurality of above-described radiation detector units **210** mounted to an above-described frame bar **140**. A column of radiation detector units **210** may be mounted to the front side of the frame bar **140**. Engagement

features **214** may optionally be provided on the front side of the frame bar **140** that may mate with corresponding engagement features (not shown) on the backside of the supporting substrate (e.g., carrier board **60**) of each of the radiation detector units **210**. End holders **260** may optionally be located at either end of the column of radiation detector units **210**. A module circuit board **220** may be mechanically coupled to the frame bar **140** by suitable mechanical fastener(s). Each radiation detector unit **210** of the detector module may be electrically coupled to the module circuit board **220** by a flex cable assembly **62**. In the embodiment shown in FIG. 4, the module circuit board **220** may include board-side connectors **212**, where each board-side connector **212** may be connected to a connector, such as a snap-in connector **66**, of a respective flex cable. In the embodiment shown in FIG. 4, each detector module **200** includes a column of eight (8) radiation detector units **210**, each including a pair of radiation sensors **80** mounted over an ASIC **130**. It will be understood that other configurations for the detector modules **200** are within the scope of this disclosure.

[0055] FIG. 5 is a top view of a detector array **300** including a plurality of detector modules **210** according to comparative embodiments of the present disclosure. Referring to FIG. 5, the detector array **300** may include a plurality of above-described detector modules **200** mounted on a detector array frame **310** to provide a continuous detector surface. For clarity of illustration, adjacent detector modules **200** are depicted in alternating shading in FIG. 5. The detector surface may include an $m \times n$ array of pixel detectors **126** including n rows of pixel detectors **126** extending along the Φ direction and m columns of pixel detectors **126** extending along the Z-axis direction, where m and n are integers.

[0056] Each detector module **200** of the detector array **300** may include a plurality of radiation detector units **210** mounted on a common support. In the embodiment shown in FIG. 5, each module includes a column of eight (8) radiation detector units **210** extending along the Z-axis dimension, where each radiation detector unit **210** includes a pair of radiation sensors **80** located adjacent to one another along the Φ direction and mounted over a common ASIC **130**. Other suitable configurations for the detector modules **200** and/or radiation detector units **210** of the detector array **300** may also be utilized.

[0057] Radiation sensors **80** as described above are typically sourced from a wafer of a radiation-sensitive semiconductor material (e.g., CZT) that is singulated (i.e., diced) to provide a plurality of individual radiation sensors **80** or “tiles.” Accordingly, pixel detectors **126** located on the edges of the detector may be sensitive to defects created at the crystal (e.g., tile) edge during dicing. These defects may create increased surface currents that flow around the corners and edges of the detector tile and may increase charge generation, which can cause premature reverse breakdown in the detector tiles which are configured as Schottky diodes, as well as excess currents at operating voltages. Defects arising from cutting may also create a high-density of carrier traps with long de-trapping time and may result in ballistic deficit and reduction in charge collection efficiency of the edge pixel detectors **126**. Defects arising from cutting or due to surface contamination may disturb the electric field in the sensor such that photo generated carriers are directed to an electrode that is not aligned with the location (voxel) where the radiation was absorbed.

[0058] Thus, the pixel detectors **126** along the peripheral edges of radiation sensors **80** tend to exhibit relatively poorer performance and reliability than the pixel detectors **126** located in the interior regions of the radiation sensors **80**. The presence of these edge pixel detectors **126** may be particularly problematic for CT reconstruction because the image data is typically processed in “slices,” meaning that image data from each row of pixel detectors **126** located at the same position along the Z-axis direction may be processed together. Because the edge pixel detectors **126** in the detector array **300** shown in FIG. 5 are all aligned, this means that a number of rows in the detector array **300** are composed entirely of edge pixel detectors **126**. One such edge pixel detector **126** row, n -x, where x is an integer less than n , is shown in FIG. 5. In the detector array **500** shown in FIG. 5, for example, sixteen (16) of the n rows of pixel detectors **126** are edge pixel detector rows made up entirely of relatively poor performing edge pixel detectors **126**. Moreover, many of these edge pixel

detector rows occur consecutively along the Z-axis direction between adjacent rows of radiation detector units **210**, which may further negatively affect image quality.

[0059] It may be possible to improve the performance and reliability of edge pixel detectors **126** by providing a protective coating over the sidewalls of the radiation sensors **80**. The protective coating may prevent defects on the sidewalls that reduce the performance and reliability of the edge pixel detectors. However, providing a protective coating over individual radiation sensors **80** can add complexity and cost to the manufacturing process. In particular, the handling of individual radiation sensors **80** can be challenging. The coating process may further require masking of both the cathode and anode electrode surfaces to prevent deposition and/or separate etching processes to remove the coating from select regions of the radiation sensors **80**. The masking can be very difficult around edges and corners.

[0060] Various embodiments include using a conformal deposition process to form a protective coating over the side walls of the radiation sensors **80** after the radiation sensors **80** have been assembled into an above-described radiation detector unit **210**. In accordance with various embodiments, one or more radiation sensors **80** may be mounted over the front side of an ASIC **130**. The radiation sensors **80** may be mounted over the ASIC **130** via an interposer **40**, such as shown in FIG. 3A, or via a direct attach method as shown in FIG. 3B. One or more ASICs **130** having radiation sensor(s) **80** mounted thereto may be provided on a carrier board **60** to provide a radiation detector unit **210**. The assembled radiation detector unit **210** may be introduced into the reaction chamber of a conformal deposition system. In some embodiments, the conformal deposition system may be an atomic layer deposition (ALD) system. However, other suitable conformal deposition systems, such as a chemical vapor deposition (CVD) system may also be utilized. The deposition system may be used to conformally deposit a protective layer over exposed surfaces of the assembled radiation detector unit **210**, including over sidewalls of the radiation sensors **80** of the radiation detector unit **210**. In one embodiment, the protective layer may be deposited by ALD at a temperature of 200° C. or less, such as 100 to 200° C.

[0061] Various embodiments provide a straightforward and cost-effective method for providing a protective coating over the sidewalls of the radiation sensors **80** of a radiation detector unit **210**. By forming the protective coating after the radiation detector unit **210** is assembled (e.g., after the radiation sensor **80** is electrically connected to the ASIC **130**), additional handling and/or processing of the individual radiation sensors **80** may be avoided and standard assembly processes for the radiation detector unit **210** may be utilized. The assembled radiation detector units **210** may undergo testing/validation procedures prior to the coating process, such that only known good radiation detector units **210** may be coated. Because the radiation detector units **210** may be coated using a conformal deposition process, all exposed surfaces of the radiation detector units **210**, such as surfaces of the carrier board **60**, the ASIC **130**, and the bonding material portions **82**, **52**, **192** (e.g., solder balls, conductive epoxy, etc.), may also be coated by the protective coating. The protective coating may be composed of a suitable electrically insulating material that does not interfere with or otherwise affect the functionality of these components. Furthermore, by performing the conformal coating process after the radiation detector units **210** are assembled, the requirements for masking portions of the radiation sensors **80** may be reduced. In some embodiments, only select regions of the radiation detector unit **210**, such as the areas corresponding to the cathode electrode **122** and the connector for the radiation detector unit, may be masked.

[0062] FIGS. 6A-6D are side cross-section views illustrating a process of forming a protective coating **401** over exposed surfaces of a radiation detector unit **210**, including over the sidewalls **403** of the radiation sensors **80** according to various embodiments of the present disclosure. Referring to FIG. 6A, an above-described radiation detector unit **210** including a carrier board **60**, an ASIC **130** mounted over the front side of the carrier board **60**, and a plurality of radiation sensors **80** mounted over the front side of the ASIC **130** is illustrated. Each radiation sensor **80** includes a front (i.e., top) side **121** containing the cathode electrode **122** thereon, and a back (i.e., bottom) side **127**

facing the ASIC and containing a plurality of anode electrodes **128** thereon. Although the radiation detector unit **210** of FIG. **6A** is a “direct attach” type radiation detector unit **210** that does not include an interposer **40** located between the radiation sensors **80** and the ASIC **130**, it will be understood that the process steps shown in FIGS. **6A-6D** may also be utilized for a radiation detector unit **210** that includes an interposer **40** shown in FIG. **3A**. As shown in FIG. **6A**, the radiation sensors **80** of the radiation detector unit **210** may have exposed sidewalls **403** around the periphery of the radiation sensors **80**. In one embodiment shown in FIG. **6E**, the radiation sensor **80** may include an anode encapsulation layer **129** that partially covers the metal regions of the pixels (i.e., the edge portions of the anode electrodes **128**) and covers the region (e.g., “streets”) between the metal regions (i.e., the anode electrodes **128**) of the pixels. This anode encapsulation **129** layer may be 1 nm to 5 microns thick and may be made from any suitable insulating organic or inorganic materials.

[0063] FIG. **6B** is a side cross-section view of the radiation detector unit **210** with a mask **405** formed or positioned over select portions of the radiation detector unit **210** according to various embodiments of the present disclosure. Referring to FIG. **6B**, the mask **405** may provide a physical barrier between select surface region(s) of the radiation detector unit **210** and process gas(es) used during a subsequent deposition process such that a protective coating **401** is prevented from forming over the region(s) protected by the mask **405**. In some embodiments, the radiation detector unit **210** may be provided on a fixture or have tape attached that may be configured to physically block certain surface region(s) of the radiation detector unit **210** from being exposed to process gas(es) during the subsequent deposition step. In other embodiments, an organic mask layer, such as an epoxy or polymer material, may be deposited over the cathode electrodes **122** and the connector for the cable assembly **62** to form the mask **405**. For example, the mask **405** may be a photolithographically-formed mask **405** that may be formed by depositing a photosensitive material (e.g., a photoresist) over the radiation detector structure **210** and exposing select portions of the photosensitive material to radiation through a patterned mask such that the portions of the photosensitive material that are exposed to the radiation become either more or less soluble than the unexposed portions of the photosensitive material. Then, a development process may be utilized to remove the relatively more soluble portions of the photosensitive material, leaving a patterned mask **405** over select regions of the radiation detector structure **210** as shown in FIG. **6B**. It will be understood that other suitable techniques for providing a mask **405** as shown in FIG. **6B** may also be utilized.

[0064] Referring again to FIG. **6B**, in various embodiments, the mask **405** may be located over the cathode electrodes **122** of each of the radiation sensors **80** of the radiation detector unit **210**. The mask **405** may also cover other regions of the radiation detector unit **210**, such as the connector for an above-described cable assembly **62** that may be utilized for transmission of power and data signals to and from radiation detector unit **210** in an assembled detector module **200** and/or detector array **300**.

[0065] FIG. **6C** is a side cross-section view of the radiation detector unit **210** with a protective coating **401** deposited over exposed surfaces of a radiation detector unit **210** according to various embodiments of the present disclosure. Referring to FIG. **6C**, the radiation detector unit **210** may be provided within the reaction chamber **410** of a conformal deposition system. In some embodiments, the conformal deposition system may be an atomic layer deposition (ALD) system. Other suitable conformal deposition systems may also be utilized. The deposition system may be operated to introduce one or more process gases, such as a sequence of ALD precursor pulses, into the reaction chamber **410**. The process gases introduced into the reaction chamber **410** may react to deposit a protective coating **401** over exposed surfaces of the radiation detector unit **210**. In various embodiments, the protective coating **401** may be conformally deposited over all surfaces of the radiation detector unit **210** that are not covered by the mask **405** or are not otherwise physically shielded from the process gases. As shown in FIG. **6C**, for example, the protective coating **401** may

be deposited over the sidewalls **403** and lower surfaces of the radiation sensors **80**, over external portions of the anode electrodes **128**, the bonding material portions **82** between the anode electrodes **128** and the ASIC **130**, the upper, lower and side surfaces of the ASIC **130**, the bonding material portions **192** between the ASIC **130** and the carrier board **60**, and the upper, lower and side surfaces of the carrier board **60**. In embodiments in which a mask **405** is provided over the cathode electrodes **122**, the protective coating **401** may also be formed over the mask **405**. If the anode encapsulation layer **129** is located over the backside of the radiation sensor **80**, then the protective coating **401** may be formed over the anode encapsulation layer.

[0066] In various embodiments, the protective coating **401** may be formed of a suitable electrically insulating material. In some embodiments, the protective coating **401** may be composed of one or more inorganic materials (i.e., a material that does not include carbon-hydrogen bonds). In one non-limiting embodiment, the protective coating **401** may include aluminum oxide (Al.sub.2O.sub.3) or titanium oxide (TiO.sub.2), or mixtures (e.g., aluminum titanium oxide) or nanolaminates thereof. Other suitable materials, such as any insulating metal oxide, metal nitride or metal sulfide, or semiconductor oxide or nitride (e.g., silicon oxide, germanium oxide, silicon nitride or germanium nitride) or mixtures (yttrium zirconium oxide, silicon oxynitride, etc.) or nanolaminates thereof may be used. For example, insulating materials include scandia, yttria, zirconia, hafnia, niobium oxide, tantalum oxide, aluminum nitride, etc. As used herein, a nanolaminate includes at least two insulating layers having different compositions from each other and thicknesses of 10 nm or less, such as 0.4 to 5 nm each.

[0067] Other suitable materials for the protective coating **401** are within the contemplated scope of the disclosure. In various embodiments, the protective coating **401** may have a thickness of 150 nm or less, such as between about 2 nm and about 100 nm (e.g., 40-60 nm), including about 50 nm. In some embodiments, the deposition of the protective coating **401** may be performed at a temperature that is less than about 200° C., such as between about 20° C. and about 150° C., to avoid damaging the carrier board **60**. In other embodiments, the deposition may be performed at a higher temperature. Preferably, the deposition temperature is below a temperature at which the carrier board **60** is damaged and/or the solder balls are reflowed.

[0068] FIG. **6D** is a side cross-section view of the radiation detector unit **210** with a protective coating **401** following the removal of the mask **405** according to various embodiments of the present disclosure. Referring to FIG. **6D**, the radiation detector unit **210** may be removed from the reaction chamber **410** of the deposition system. In embodiments in which a mask **405** is utilized, the mask **405** may be removed via a suitable process, such as by selective dissolution with a solvent. In some embodiments, a liftoff process may be utilized to remove the mask **405** in addition to any of the protective coating **401** that is deposited over the mask **405**. As shown in FIG. **6D**, select regions of the radiation detector unit **210**, including the cathode electrodes **122** and optionally connector(s) **62** for a cable assembly (not shown in FIG. **6D**), may be exposed through openings the protective coating **401** (i.e., are not covered by the protective coating). In various embodiments, portion(s) of the radiation detector unit **210** may be exposed through the openings in the protective coating **401** without requiring an etching process to etch portions of the protective coating **401**.

[0069] FIG. **7** is a side cross-section view of the radiation detector unit **210** with a protective coating **401** according to another embodiment of the present disclosure. The radiation detector unit **210** shown in FIG. **7** may be similar to the radiation detector unit **210** described above with reference to FIG. **3A**. In particular, the radiation detector unit **210** may include an interposer **40** located between the radiation sensors **80** and an ASIC **130** as described above with reference to FIG. **3A**. The assembled radiation detector unit **210** may be placed in a reaction chamber of a conformal deposition system as described above, and a protective coating **401** may be formed over exposed surfaces of the radiation detector unit **210**. In the embodiment shown in FIG. **7**, the protective coating **401** may be deposited over the sidewalls **403** and lower surfaces of the radiation

sensors **80**, over portions of the anode electrodes **128**, the bonding material portions **82** between the anode electrodes **128** and the interposer **40**, the upper, lower and side surfaces of the interposer **40**, the upper and side surfaces of the ASIC **130**, the solder balls **52** extending between the interposer **40** and the carrier board **60**, and the upper, lower and side surfaces of the carrier board **60**. In the embodiment, an insulating matrix **34** may be located around the array of first bonding structures **32** between the interposer **40** and the ASIC **130**. The protective coating **401** may be located over the surface of the insulating matrix **34**.

[0070] FIG. **8** is a flow diagram illustrating a method **600** of fabricating a detector structure, such as an above-described radiation detector unit **210**, according to various embodiments of the present disclosure. In step **601** of method **600**, a detector structure **210** may be assembled by mounting at least one ASIC **130** over the front side of a carrier substrate **60** and at least one radiation sensor **80** over the front side of the at least one ASIC **130**. In step **602** of method **600**, a protective coating **401** may be formed over at least a portion of the assembled detector structure **210** including over sidewalls of the at least one radiation sensor **80**.

[0071] The devices of the embodiments of the present disclosure can be employed in various radiation detection systems including computed tomography (CT) imaging systems. Any direct conversion radiation sensors may be employed such as radiation sensors employing Si, Ge, GaAs, CdTe, CdZnTe, and/or other similar semiconductor materials.

[0072] The radiation detectors of the present embodiments may be used for medical imaging as radiation detectors in High-Flux applications as in X-ray Computed Tomography (CT) for medical applications, and for non-medical imaging applications, such as in baggage security scanning and industrial inspection applications.

[0073] While the disclosure has been described in terms of specific embodiments, it is evident in view of the foregoing description that numerous alternatives, modifications and variations will be apparent to those skilled in the art. Each of the embodiments described herein can be implemented individually or in combination with any other embodiment unless expressly stated otherwise or clearly incompatible. Accordingly, the disclosure is intended to encompass all such alternatives, modifications and variations which fall within the scope and spirit of the disclosure and the following claims.

Claims

1. A detector structure, comprising: a carrier board; at least one application specific integrated circuit (ASIC) located over the carrier board, the at least one ASIC comprising signal processing channel circuitry; at least one radiation sensor located over a front side of the at least one ASIC, the at least one radiation sensor having a front side and a back side, wherein the back side of the at least one radiation sensor faces the front side of the at least one ASIC; and a protective coating located over sidewalls and at least a portion of the back side of the at least one radiation sensor and over at least a portion of the at least one ASIC and over at least a portion of the carrier board.
2. The detector structure of claim 1, wherein the protective coating has a thickness between 2 nm and 100 nm.
3. The detector structure of claim 2, wherein the protective coating has a thickness between 40 nm and 60 nm.
4. The detector structure of claim 1, wherein the protective coating comprises an inorganic material.
5. The detector structure of claim 4, wherein the protective coating comprises an insulating metal oxide, metal nitride, metal sulfide, semiconductor oxide or semiconductor nitride, or mixtures or nanolaminates thereof.
6. The detector structure of claim 5, wherein the protective coating comprises aluminum oxide or titanium oxide or a mixture or a nanolaminate thereof.

7. The detector structure of claim 1, wherein each radiation sensor comprises a cathode electrode on a front side of the radiation sensor, wherein the cathode electrode is exposed through an opening in the protective coating.
 8. The detector structure of claim 7, wherein each radiation sensor comprises a plurality of anode electrodes on a back side of the radiation sensor, wherein the protective coating is located over a portion of each anode electrode.
 9. The detector structure of claim 7, further comprising a plurality of first bonding material portions located between anode electrodes of the at least one radiation sensor and the at least one ASIC, wherein the protective coating is located over a portion of the first bonding material portions.
 10. The detector structure of claim 7, further comprising a plurality of second bonding material portions located between the at least one ASIC and the carrier board, wherein the protective coating is located over a portion of the second bonding material portions.
 11. The detector structure of claim 1, wherein the at least one radiation sensor comprises cadmium zinc telluride or cadmium telluride.
 12. The detector structure of claim 1, wherein the front side of the at least one ASIC is bonded to the back side of the at least one radiation sensor.
 13. The detector structure of claim 1, further comprising at least one interposer located between the front side of the at least one ASIC and the bottom side of the at least one radiation sensor.
 14. An X-ray imaging system, comprising: a radiation source configured to emit X-rays; and a detector array including a plurality of detector structures of claim 1 that form a continuous detector surface and that are configured to receive the X-rays from the radiation source through an intervening space configured to contain an object therein.
 15. A method of fabricating a detector structure, comprising: assembling a detector structure by mounting at least one application-specific integrated circuit (ASIC) over a front side of a carrier substrate and at least one radiation sensor over a front side of the at least one ASIC; and forming a protective coating over at least a portion of the assembled detector structure including over sidewalls of the at least one radiation sensor.
 16. The method of claim 15, wherein the step of forming the protective coating comprises placing the assembled detector structure within a reaction chamber of a deposition apparatus after the step of assembling the detector structure, and introducing one or more process gasses into the reaction chamber to conformally deposit the protective coating over exposed surfaces of the assembled detector structure.
 17. The method of claim 16, wherein the protective coating comprises an inorganic protective coating that is formed via atomic layer deposition (ALD) at a temperature of less than 200° C.
 18. The method of claim 15, further comprising providing a physical barrier over one or more select portions of the detector structure to prevent deposition of the protective coating over the one or more select portions of the radiation structure.
 19. The method of claim 18, wherein providing the physical barrier comprises forming a mask over the one or more select portions of the radiation structure.
 20. The method of claim 18, wherein the one or more select portions of the detector structure comprises at least one cathode electrode on a front side of the at least one radiation sensor.
 21. The method of claim 18, wherein the one or more select portions of the detector structure comprises at least one connector for a cable assembly.
 22. The method of claim 15, wherein the protective coating comprises an insulating metal oxide, metal nitride, metal sulfide, semiconductor oxide or semiconductor nitride, or mixtures or nanolaminates thereof.
 23. The method of claim 22, wherein the protective coating comprises aluminum oxide or titanium oxide, or mixture or nanolaminate thereof.
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